4 – October Focus Group – FG3

Interviewer

We're going to start off by talking about blood glucose profiles. And so what I mean by that is kind of that view often when you look at a CGM of some kind and just that kind of graph that shows the changing levels. And we'll talk a bit about kind of what levels you like to keep or aim for and how that changes and what activities or food do to that. So yeah, you should just feel free to edit this at any point, whether it's kind of draw a line or kind of notes and jot some things down. But it could be really rough or anything it's just sort of to help get your points across. Actually, I will shut that for now because we quickly go through the instructions just because obviously these focus groups change every time. As previously we'll just go through, say what type one diabetes technology you're using, which smartwatch you're using as part of this study. And then this time, it'll be either a little gadget you've found helped with your type one diabetes management or a little right bit of advice you might find helpful or you kind of figured out. And that could be really anything, it's just maybe a quick point if you have one. So, I'll start off. Um. Yeah, as I've said a few times already, I'm now on the t:slim and Dexcom G6 sensor. I had on the, in fact on the t:slim at the weekend, I had a bit of a scary moment when it just stopped working altogether and said like operations failure. Um. Which was a bit scary, but I guess shift back to the pens. It's good that you know how to use those and now it seems to have sorted itself out. But, it does really show you those moments when technology begins to break down you have to resort back. Um. I'm using a Fitbit sense just kind of part of PhD to get an idea of how they work and function. And the little hack I found, in the last few years has really helped me out, is just around using activity to kind of speed up the effect insulin has. Whether that's, yeah, usually kind of just a walk, if I'm really struggling to get them to drop, just it really helps push those sugars down a bit further. It probably has taken a bit too far at points, I definitely can remember running on the spot in the flat just to kind of drive them down at particular times. Um. But yeah. Otherwise, that's yeah kind of everything, so [P06] do you mind going next and introducing yourself?

P06

Hi so I'm on the Omnipod and I'm using Libre FreeStyle 2. Um. I've got the Fitbit Luxe. Um. I'm not sure this is advice, but something I've been sort of struggling with at the moment is dealing with pasta. I seem to eat quite a lot of that and I seem to wake up high most nights if I forget how to deal with it, but yeah. So watch out for that.

Interviewer

Yeah, I think everyone's had to deal with that some stage or another and it is an ongoing fight. Would you mind picking someone to go next as well it should?

P06

Um. [P03].

P03

Uh, yeah. So, I'm on the uh Tandem t:slim and the Dexcom G6 and I have Fitbit Sense. And mine is similar to [Interviewer] in that hot water also helps increase your insulin. So, if I'm really struggling, I have a hot shower or a hot bath. But on the flip side I have I have also gone low in the shower many times.

Interviewer

Yeah, that's interesting the little tricks you might find. I hadn't heard of that one.

P03

And [P13], do you want to go next? You're next on my screen. I think you're on mute.

P12

Sorry. Yeah. I'm. I'm on the Medtronic 780G and the Guardian 4 sensor. I've got the um Fitbit Versa 4 and my hack I've found, which makes me look really weird, but I don't care, squat help me bring my blood sugar down. So, there's been many times where I'm at work and I've been running high and I've just had to go right "I'm just gonna go on the cellar for a minute", and I'm just like doing squats in the cellar. And people have walked in and they're just like, "what are you doing?" "Like I'm high. Just leave me alone. I'll be back in a minute." But yeah, they they help my insulin just kick in a little bit better and yeah, start bringing it down.

Interviewer

Yeah, really interesting that kind of how different activities can do different things and and you found that one works for you. Uh, you know, I I think that just leaves you, [P21], if you're happy to go?

P21

Yeah, I'm [P21]. I'm on the Omnipod and the FreeStyle Libre 2 and I've got the Fitbit Luxe and I don't know what my tip would be. I guess one that I've been doing recently is before exercise or running or something, I used to always sort of take on board carbohydrates 40 or so minutes beforehand. But recently I've been suspending my insulin about about that time beforehand instead, which I found has has been working and to stop me going from going low when exercising.

Interviewer

Yeah, no, great. It is. I think there's with all these things, it's that trial and error process of finding what works for you. Oh brilliant. So yeah, as I was saying, we kind of kick off and uh the first point is um I want to talk or hear from you a little bit about on is around this idea of what your optimal BG

might be. Or what the blood glucose you might aim for is and how they kind of compare at different times. If anyone's just happy to start on that.

P06

Do you have any like specific time in mind or are you just talking across the whole day?

Interviewer

Across the whole day, I think. Um. Yeah, this really works best if you just kind of pick a point that you, anything to do with it and then people can kind of chip in on that if they agree, disagree or at different times do different things.

P06

Right. Well, I I'd say I aim to keep my blood sugar between 4 and 8 kind of during normal times. But then if I eat it kind of spikes up to 10, that's I'm OK with that. I think it starts going above that I try and keep an eye on it and try and reduce that. But sometimes, yeah you can't help it.

Interviewer

Yeah. Is that similar for other people? Does anyone have um different time or?

P03

Yeah, that's kind of similar for me.

P21

Yeah. For me, I think I'm a bit worried about overnight going low so I I I tend to find that I'll run myself slightly higher going to bed than I would do if I was in the days sort of actively checking checking things more probably just buy 1 or 2 mmol/L. I'll sort of go to bed a little bit higher.

P12

I'm like that, but at work as well. I don't mind running like a tiny bit higher when I'm at work, just so that I know I'm not going to have a hypo while I'm there because I'd rather just be high and have to deal with just that not feeling amazing then and being low and then literally not being able to actually do my job.

P21

Yeah.

Interviewer

Yeah. So, it seems across all you've been saying, there is kind of maybe this typical target range. Yeah, there's someone who's down there this 5 and 8. But then there are occasions where it shifts. So, we've heard overnight, after eating, at work. Are there any other times that you kind of would look to maybe maintain your blood glucose levels that are different points to usual?

P06

Uh, when I'm driving, I actually try and keep it higher uh, because I can't check it all the time. So, I kind of prefer to have it just a bit higher than risk having to stop for a while.

P03

This is a really, really random situation. I've got an MRI scan in about 10 days and I'm probably going to try and have myself run a bit higher because I can't have my pump on my Dexcom whilst I'm in there. So, I'm probably going to beat myself, run higher so I don't have to worry about crashing low either in the MRI scan body machine thing or the 2 hour warm up period after. So, it'll be like 3 hours when I'll be without the Dexcom.

Interviewer

Yeah, they these will seem like, uh, yeah, sensible reasoning points. UM, I think it's interesting that all these points are you look for an increase in blood glucose or look to run slightly higher, mostly to avoid typos I guess. Are are there any times where you try and maintain your blood glucose a bit lower for any reason?

P03

I guess if I was going to like a all you can eat buffet, I'd want to be a bit lower when I'm going in. Because just cause I don't trust my estimation and I know I'd run the risk of going high. So, it's just to kind of cut out as much of going as high.

Interviewer

Yeah, yeah. Anyone else got any ideas?

P03

Or maybe like pizza, cause pizza tends to just send me uh sky high.

P06

Uh, normally, before I well, if I'm going drinking, then I only run myself a bit lower before that because I always get that initial spike but it does seem to drop off at the end, so I yeah, you gotta be careful about giving too much or like doing it too late.

Interviewer

No, definitely. And I guess across across the yeah, participants we've got here, there is like a range of different technologies using with some on closed-loop systems and some not. Yeah. How do you feel about the role technology currently plays for you in like changing this target level and is there anything extra you'd want in any way?

P03

I think for me, anything that takes as much of the thinking out of like my management. Like I still think that I should have some input. But like I had my HbA1c today and it's like being the lowest I've ever had and I've that's because of the technology. And I've just been like I was like, really worried it was gonna be super high, but it was quite it was quite a good HbA1c. Um. And it's like we get, we see so much of like the fine details because of technology. Yeah. Anything where it's like removing as much of our thoughts out of it and still getting like a really good result would be beneficial.

Interviewer

Yeah, totally. Um. And thinking maybe then as well, thinking more specifically at those times, so everyone outlined quite a few occasions where they try and maintain their blood glucose, little higher. How would you go about doing that?

P06

Uh, well, I go about by putting myself on temporary basal rate that's a bit lower than my usual one. Unless I need a really quick thing, in which case I'll eat something that will send me up. But those are the two main ways I'd do it really.

P03

Yeah, I'd echo that.

P21

Yeah, same here or suspend for however long.

Interviewer

And what about you, [P12]? What do you do?

P12

Yeah. No mines yeah, pretty much the same. Just it's usually more suspending than using temporary basals just because I don't always I don't fully trust the temporary basal. I've I've tried them in the past, they haven't really they've not done what I've wanted them to do. So, I'd rather just suspend, do what I need to do and then just unsuspend it.

Interviewer

Right. Yeah, it's OK. So, a lot of people talk about the temporary rates and reducing it that way. I guess with the with closed loop systems, there's tends to be an actual target the system's aiming for, and often there's a way of changing that, whether that's with activity features or I think they're commonly called or just increasing that for a while. Has anyone on those systems used those or? I mean, so [P03], you said you were on the the t:slim, that’s right?

P03

Yeah

Interviewer

So, I I know particularly that one has the activity feature. Is that something you ever use or?

P03

It's not, no. I think I used a sleep feature but I'm quite a forgetful person, so I know if I was to put it in activity mode, I'd forget to put it back into normal mode. Whereas with the sleep one, I just have a. I can't remember what it what I've said it up as. I think it's like 11:00 till 7:00 for like a sleep, sleep routine.

Interviewer

Right. Interesting. So is there any way you think the the activity feature could be changed to make you more likely to use it?

P03

Maybe. I I don't know if this actually is a feature, but I don't think it is. If it was like, how long do you want to have this activity mode enabled. So, if you know that it's going to be like a a couple of hours you can just say 120 minutes or as long or as little as you need.

Interviewer

Right, definitely, yeah that makes a lot of sense. I I guess then [P12] on the MiniMed system, do they have any features similar to that? My awareness is less.

P12

Not really. I mean you can do like like um preset um like um temporary basals like you can just do them where it like. I'm looking at it now. Where it just puts in a certain amount like per hour and and you can change it depending on whether or you think you do want like high activity or moderate activity. But you you have to like set those limits for yourself. Um. But we don't really have like a proper like activity one or like a sleep one or anything like that. It's um yeah, it's just those preset ones, really.

Interviewer

Yeah. Is that's something you use ever?

P12

Um. They don't. I've tried them and I've got I've got ones on there for like high activity, moderate activity, but I don't ever change them um just because I think, I don't know. I don't. Part of me thinks I'll just forget to change it back and part of me is just like I don't know whether I fully trust it to do what I want it to do. Because it was actually the the consultant at the hospital who set them up. So, I think I'm a bit wary of using them just in case something goes wrong or um, it just doesn't do what I want it to do.

Interviewer

Yeah, totally. Do you think they could be changed in any way to make them more appealing or you more likely to use them?

P12

And maybe if it was like if it had a specific name for it instead, like maybe like like an activity mode or like a sports mode or sleep mode, things like that. If they had those kind of names, maybe I'd be more like “Oh yeah”. I can, if I check like my own like trends like overnight for sleep mode like they're. Like this is maybe how I wanted to go in these step times when I'm sleep. But yeah, if it had names like that, it might be more appealing and it might be more likely to use them.

Interviewer

Right, definitely. And I guess then [P06] and [P21], you've not gone to a closed-loop system or if you were to move on to one them, how do the idea of these kind of feature sounds? Is that something you'd want to use or is there something you'd want to use instead?

P06

Um. Yeah. So, if I were to move on to one, I think I'd give it a try. Definitely the sleep setting, that is something that makes me a bit anxious just going hypo. So, that seems like quite a good idea. And then the activity mode might be good, but it depends how much difference it actually makes. I don't seem to struggle too much doing activity, like keeping reasonable control. So, yeah, I don't know if it would be worth it to try it. Yeah.

P21

Ohh sorry, on mute. Yeah, I think similar for me. I would definitely try them out. Sorry just to clarify, is it with the activity mode, does that mean it sort is is like an automatic temporary basal depending on your levels or?

Interviewer

So I think, I guess it works a little different for closed-loops. So, they're what they're normally doing is trying to maintain you at a certain level. So, on the t:slim for example I've mine set at 6.1 or 6.3 and so it's changing the insulin dosage I'm getting to try and keep me at that level. But, then if I put it in activity mode, it suddenly aims for 9 instead, so it's gonna give me less.

P21

OK. Yeah, I think yeah, that would be something that I would find quite beneficial I think because I mean I guess like when I go to the gym or out for like a run around the park I in my head try and keep myself about 9 or 10 because I know if I go at 5 or 6, then I'll drop down. So, I guess if if there was a a closed-loop automatic way of doing that, then I would definitely give it a go and see sort of how it went.

Interviewer

Yeah. No and um. I guess I maybe going back a little, one, one thing [P03] you mentioned was around recent having an HbA1c done and I think what's interesting is quite often when you go to the hospital, there is kind of this focus on this number of HbA1c or maybe now timing range is coming in to be spoken out a bit more. How do you feel about that kind of being a reflection of your overall blood glucose management. Is there anything else you think should be considered, or do you think it is a good reflection of your kind of management management over that period?

P21

I mean, I guess they've got to measure it somehow um to get to some sort of average or or look at your data, but I don't think that HbA1c gives you the full picture at all. Because for example, somebody that has a lot of hypos and is hyper um multiple times a day would have a good a, a lower HbA1c even though their control or management isn't necessarily better. So, I think maybe looking at like number of highs, number of lows, mean or average blood sugar as well as the HbA1c would maybe be useful in appointment.

P03

Yeah, I'd agree. I think it it's a good starting point. But it definitely needs to take into consideration how much fluctuation that is in someone's blood sugars. And if they are like sitting at 3.5 most of the time. Well, not most of the time but a lot and then having like have having some really big highs that then pull the HbA1c up into like the target.

P12

I think as well the um, when you just look at the A1c, like it's just a number, it doesn't take into account things like sick days, where your bloods are probably going to be running more high than usual or things like periods which can affect your blood sugar a lot. Um. Basically it just doesn't it doesn't take into account like personal things that have gone on in that last three months. It just says what your blood sugars have been and it gives you the average. That's it. It doesn't take into account all these other things that could have been happening to you. So, like it's a it's a start, but it. Things need to be a bit more, if they could be a bit more personalised. I don't know how they would do that, but I'm not I'm not the sciency person. I can't be like, "oh yeah, this is exactly how this should do it, this, this and this". But it's not personalised enough to give a true reflection of what someone's blood sugars have been over that past three months.

Interviewer

Yeah, definitely. I think that's a really interesting point is that, yeah, does anyone else have suggestions for things they might want to be considered beyond just yeah, the value how your blood glucose values have been?

P06

Uh, pod malfunctions. Semi regularly I seem to get like a leaky pod or something and so my blood sugar goes really high. That's kind of frustrating when I know it's not going to be reflected that my my pod broke but yeah.

Interviewer

Yeah. Anyone else? I think maybe, what's come up in previous ones? Some people have spoken about, or from the point of view of like the effort that's had to gone into that management or you

know how, how you've been kind of feeling about that month of management beyond just how your blood glucose levels being. How how do you all feel about that balance?

P12

I mean, yeah, I think obviously we're all we're all trying to keep our blood sugars in a good range. We don't want to have loads of highs and loads of loads, we know the effect that they can have on us, not just like in that moment, but all the time as well. So like we are all trying our best, but that isn't again that isn't shown in your A1c. They don't know how you've been feeling over those past months either. Like you could have been having really low months, and your management's kind of gone out the window just because of how you're feeling. And because I've I know I've definitely had that in the past and it it reflects in your A1c. It was obviously higher because you've not really been trying to manage it that well. But again, they don't know what your mental state has been like. So, how are they, how are they supposed to incorporate that in. It's, yeah, that's that's I think that's that's something they should really take into account as well.

Interviewer

Yeah. Yeah. And does that? I guess that we've kind of been talking a lot about from a hospital point of view and those appointments, what they're kind of looking for or viewing in terms of your type one diabetes management. What what about more personally do you have goals for your diabetes management in some form that differ from that?

P03

I think for me my goals differ depending on how I'm feeling. Like sometimes it will literally just be have you made it to the end of the day? Have you made it to the end of the week? Well done, you survived diabetes. And I think that's like obviously the ultimate goal for for everyone. Making it through the day. And then I think sometimes if I'm in a better place, I tend to like have a target for timing range or uh a limit of don't go above this today. Um. But I just try to be gentle with myself and goals that I'm setting.

Interviewer

Yeah, yeah. What about everyone else? Is that something they kind of experience themselves or do they experience something different?

P06

Yeah, I'd can relate to that. Uh, it's more just like long term, try yeah, just keep myself healthy um for as long as possible I guess.

P12

I try not to give myself too many like day-to-day targets. Just just because I know what effect it will have on my um my mental health. I'm like, “oh, you've got to keep it below this”. And if I don't keep it below that in my head, will just that spinning and be like, “well, you could have been better. You could have done this. You could have done that.” So, I just know it'll make me spiral a little bit if I keep trying to make, give myself those targets. So, I think I think more long term like just as long as you keep yourself healthy, you're not high all the time, things like that. Then you know less risk of complications, better outlook for the future. That's I think that's as far as I go in terms of targets. Just so that I know that I won't overthink things all the time.

P21

Yeah, I'm the same really as [P12]. I try not to be too hard on myself, because at the end of the day, you can only do what you can do and do your best. But if I was gonna sort of measure it, and look at it I would sort of use the Libre chart I think. The sort of time in range on that and look at the variations and spikes and just try and keep it sort of as as flat as possible on a day-to-day basis, without going into too much like details on the specific data and numbers. Um. Yeah, just look at the daily graph.

Interviewer

Yeah. And I guess a few of you mentioning that there is that variation day-to-day in how much yeah, like scope or how much capacity you have really to focus on these different areas of type one diabetes management. Thinking about that and then kind of inputting information to devices, whether that's carbohydrate readings or make adjustments like TBRs, how much do you see that varies day-to-day? Maybe based on these things or based on other kind of factors that affect it.

P12

I think every day is different. Like you're going to do different things every day, whether that's just a different meal or a different activity. Um. It depends like whether you're at work or not, things like that. So, your days are always going to be different, so your your daily graphs, your daily outlooks, they're all going to be different as well. And a lot of things are going to be out of your control and so. I think when you look at your things like that, you can't beat yourself up too much about them. Um. I just I think as long as you've tried, and yeah, you might not get results you’ve been wanting or you know you were expecting from the day. But um things are unpredictable, your body's unpredictable. You can't control what's going to happen, so uh. Yeah, as long as you're trying your best, you've just you can't control what the outcomes going to be.

P06

Yeah, I I think I'd agree with that. I think when I've got this like mental space to think about it, I try and uh give more attention to it. But if truth be told, I'm not always great at doing that, especially when I've got deadlines and stuff coming up focus on diabetes just seems to go out the window a little bit. So yeah, the amount of attention it gets then is definitely less than during normal times.

P03

Yeah, I would also agree with that. When I'm, because I work full time, whenever there's like a lot of stressful stuff happening at work that takes up like a lot of my brain capacity, I noticed like diabetes goes a bit on the back burner. And I I try I try to. just focus on fighting the fire at work instead of diabetes, because then I've only got like 20% left of brain to deal with stuff. And it's, I think it's ultimately about trying to not burn myself out or spread myself too thin.

Interviewer

Yeah. And then it maybe thinking about it like from this view of kind of technology and the way it's going, increasingly it maybe asks you for more information about what you're doing or you're wanting to supply that so it can make better predictions about what your blood glucose might be, or make different decisions based upon it. Um. Yeah. How how do you feel about that extra input of information and yeah, on a day-to-day, particularly with that variation in kind of capacity for it?

P03

I guess if you could set up like a profile that has schedules in it. So, if you know that. So, for me, I work from home three days a week and work in the office two days a week and I'd probably have different profiles for office days and days I'm at home. And then maybe like a profile for when I'm going out and about, so that it's less demanding, it's more. And obviously you can go back in and change these whenever you need to, but I know for me I have like maybe my days in maybe 5 different ways. Like lazy days, days I'm probably going to be going out and seeing friends, office days, work from home and then miscellaneous, like where the days that I'm doing house chores going to the shop. So, having different profiles for those different things means it's less import overall, but you can just say “ohh, today's an office day”.

Interviewer

Yeah, and and and how about system? What what about a system then that might prompt you for more information? How would you feel about that at particular times with, with the aim, the aim of trying to improve that? And not that the response is required, it's just kind of that optional extra at times it might, you know, ask you for a bit more information.

P21

I feel like it's kind of like if I had gratification in that sense. Like it would be worth putting it in if it's going to make wiser decisions that will ultimately make your blood sugars in range more of the time.

P12

Yeah, I'd agree with that. I don't mind things where like it's asking you for that little bit more information. I've got a few apps where I have to like transfer data from one thing to another and. Just try and give you a like a roundabout view of things. And so. Yeah, I think I'm I’m the same with [P21]

there that I wouldn't mind doing that as if it's going to, like help, you know, make better predictions or like change limits about or things like that, yeah.

Interviewer

Yeah, yeah. Anyone else want to add to that?

P06

Ohh, I'd agree with that.

Interviewer

OK. And then I mean like building on this idea of the system, what if it was then to, yeah, almost give you a reward for that extra information in the way of providing you with information back saying maybe this little this extra bit of information might improve your timing range by this degree. Or it gives you information about more general trends in your life like, “oh, you've tend to be doing this at a certain time”. Would that kind of improve the likelihood of you interacting with it and giving that extra information? Or is there anything you? Right. Yeah, start with that. Would that change how you'd interact with it?

P06

Yeah, I think that would be quite good because yeah, sometimes I don't notice patterns in my life until I actually look over the data um and then you can see it. So, if it's suggesting this, that gives you like a moment to think on it and then if it is that then yeah, maybe that would help.

P21

Yeah, I'm the same. I think I'm quite a visual person. So, it's kind of like reminding me of, like, when you're driving in Google Maps and it says “you can save 5 minutes by going on this route” and like telling you that physically is quite helpful. So similarly, I think if it was to say that kind of information it would personally, like and and be an incentive to to make those changes.

P03

Yeah, same for me.

P12

Yeah, that, that's. Yeah, I'm pretty much just same there. Um. Seeing everything laid out is probably probably better for a lot of people. Not everyone, but a lot of people I think would prefer to see it

laid out like that. Just like “ohh yeah, this could help you with this. Do you want do you want to try this first?” And yeah.

Interviewer

Yeah. And maybe applying that idea to current. Yeah, I think that's quite an idea of, like, stuff being laid out for you. What? What about for? Um. Going. And actually I like the analogy that was made about Google Maps as well, and that improvement time that it's like a nice way of summarising it, kind of that little extra bit of, here's what it can do uh to make that adjustment. Yeah, thinking about laying it out and then around closed-loop systems. And that's think about closed-loop system for the point of understanding decisions it's making when it's maybe reducing insulin or increasing insulin. Do, I mean, do the people currently use closed-loop systems feel like you adequately get that to a degree that you're happy with? And then the people not currently use them if they were to shift on how much information would you want about that kind of explanation behind why a closed-loop systems made its decision it has?

P12

Um. Mine is closed-loop and I and I understand why it makes some of the decisions it does. But, for example, mine's got something called Smart Guard where it's, it automatically adjusts whether it's higher and when when it's low, it just tells you. But sometimes it can go out to Smart Guard, you have to recalibrate it with a fingerprick. And and when it's not in Smart Guard, I think it it suspends my insulin when my levels are about 6.1. And if it's, if it senses that it's going down, it'll stop the insulin like completely, even though it's still at a reasonable level, like 6.1, which is something I don't understand. I'd understand if it was maybe a little bit lower, maybe about 5 like high 4s, but when it's about 6, that's that's when I don't understand it. And I think. I think that's probably to do with what settings it's on, and I probably could change that, but I'm not sure where I need to go to change that.

Interviewer

Yeah. How does that compare to other people? What about someone who doesn't have a system currently. How much, yeah, would they want to know about it to trust what it was doing?

P06

I think I'd like to be able to see why it's made those decisions at some point, just for my own peace of mind. But uh, I think on the day-to-day I wouldn't really mind it just getting on with it. Just if it did store it. Somewhere so that I could check, if I ever did.

P21

Yeah, I think I'm the same. I wouldn't be particularly bothered to say why every decision has been made, but I would definitely like to have that sort of peace of mind that I can access that information somewhere just on the on the settings. Um, yeah.

Interviewer

Yeah, and and and on that level of yeah, explanation you want of these systems. Maybe historically on other bits of technology you've used within diabetes, has that changed over the time you've been using it? As in, how does it change the longer you use it? Do you want more or less um information from it? Or do you want it to be the same consistently through?

P21

I guess maybe it's a case of like building up trust. So maybe I feel like at first I would want more information and as I got to trust and learn how it works with my body and how it reacts to highs and lows and things then I would trust it more and um not feel like I needed to know the sort of backend behind all of those decisions.

P12

Yeah, this pumps the first pump I've ever had, like as a proper one. I trialled one when I was like, very early diagnosed and and it wasn't for me and it took another about 8 or 9 years before we actually got onto this one and I was really wary of it at first. I didn't know, you know how it was gonna work properly, how it was going to just react to, like, my body, my insulin levels, everything like that. And and I've had it about a year and a half now and as that time has gone on, obviously, yeah, I've started to trust it more. Like it knows what it's doing it. Like when to, when to step in and intervene and when it doesn't. So yeah, I think it does take that time to to warm up and trust it. And and I think at the start, I was like I need to see why it's doing this. I want to know why it's decided to stop it or why it's decided to give me some more here. Um. But now I'm just like "it knows what it's doing. I don't I don't need to see the reason why. It just knows that it needs to." And I'm fine with that.

Interviewer

Yeah, yeah, that definitely makes a lot of sense. Um. One thing as part of that as well is it's something I definitely like I found moving on to closed-loop systems has, suddenly my my management is that much better and I can maintain that level without having to do as much. Maybe I estimate my carbohydrate a bit more loosely or um I don't always tell it that I'm about to do exercise or something like that. Is that a trend anyone else has noticed, that kind of the level of or the standard of management you're getting impacts how much more effort you're willing to put in?

P12

I think I still put a lot of effort into food and carb counting, things like that just because I like it to be accurate, or at least as accurate as I can be. Obviously, there are times you have to guesstimate. I went out food last night and there's no carbs for anything, anywhere. So that was definitely a guesstimate. But, when I can carb count properly like weigh things out or if it's something set like you go on to somewhere like McDonald's and all your carb information, you can just find that out online um then I'm putting all that. Um. But obviously with the closed-loop it knows like when it's going up,

when to start giving insulin, when it's peaking and things like that and when it's settling dow. So, I don't put as much effort into like having the correct hypers or anything like that if I've messed up on food cabs and things like that, but I still put all that effort into the carbs and the food just. But that's just a personal thing, that that's not because I've been told. to um or anything else. That's just because I feel like I need to in order to have that control.

Interviewer

Yeah, yeah, that's that's really interesting to hear yeah, kind of that's perspective and it is true, it's it's personal to everyone and everyone has the the relationship with it and adjust accordingly. Yeah, does anyone have any experience that kind of shares that or that differs from that particularly?

P06

Where when I first went on to a uh continuous glucose monitor, um I wasn't great about checking my blood glucose during the day with like a fingerprick. Um. And so I wouldn't have put any more effort in, it's just that when I went on to the CGM, it was just effort I didn't need to put in. Less guilt on my mind maybe, but it felt a bit easier.

P12

Yeah, I kind of agree with that as well. Sorry. My first CGM was the first FreeStyle Libre and again, I was terrible fingerpricking before that. Um. And then when I got the FreeStyle Libre, obviously all I had to do was swipe my phone on it and it was so much easier and it felt like less effort but you were still putting the effort in. So, it kind of tricks you into making the effort, I think, but it's probably a good thing that it did. Because then obviously you can see what your glucose levels are doing instead of, you know, going for hours and hours knowing nothing.

P03

Yeah, I would agree with that actually.

Interviewer

Yeah, yeah, I I think that is really an interesting example, is that with the CGM, in fact the system allows you to put like effort in because it provides better. Can anyone think of examples in their current management that would benefit from that help or you could see being helped in a particular way to reduce that effort?

P06

I'll tell you the biggest effort I put in, is probably carb counting or like yeah, giving insulin for food. If that could be made less, yeah, easier then that would be good, but.

P03

Yeah. Yeah, that's the same for me. And like, I typically eat the same few foods, just so I can have it like in my brain. But I guess if there was like “ohh you normally have like this from a takeaway”, have this is like a preset meal. And like I guess unlimited preset meals or a high limit number of preset meals. So, like all your go tos, you can be like, “oh, I don't need to rework this out, this is what worked last time. Bang. There we go.”

P12

Yeah, I think I think the big one for me would be being able to puts insulin in from my phone. Obviously, the Medtronic ones, they don't let you do that. You've got to go into the physical pump and then put your carbs in and then it gives you the insulin. But if you're wearing something where it's not really easily accessible it's it's chew, it's a chew on sometimes they're, if I'm like, wearing a dress, I'll have to go into the toilets to give myself my insulin, which is annoying. Just because I don't want to have to rip through the dress to find the pump and accidentally like expose something that people don't want to see. Um. So, that would probably make things a lot easier. Um. And then also a bit like what [P03] said you could have like meals somewhere. Whether that's in like for for mine, maybe the Medtronic app and then you could press that. Ohh yeah this is how many carbs were in it the last time you ate it it's pretty much the same serving size just go ahead and do that again, because it worked last time and should work out the same this time as well.

P03

And I guess it's something I just added on, so add on to the preset meals if it didn't work last time, it could make recommendations on, “oh, this is you had this meal last week and you you went a bit high. We'd recommend just doing this.” Or “you went a bit low; we'd recommend doing that”. And then you can accept that and then that automatically saves into that preset meal. So then next week, if you were going. to have it again, it could be like, “OK, well, this is what we did last week. This worked. Here we go again.”

Interviewer

Yeah, I I like this process we're going on of kind of like, brainstorming an idea yeah to solve this problem around um yeah the effort around carbohydrate counting and how you deal with those meals. And yeah, there is that clear one of if we the more we can do on a phone rather than having to get a pump that's a good point. But also, yeah the system where you have preset meals. Um. Yeah, maybe to delve deep into that, is that purely you kind of build that up yourself in terms of when you have a commonly had meal, you're adding a new one. Or is it you're kind of taking from a database of existing ones or something entirely different? What would you want?

P03

I think I would personally want something where. Because a lot of meals I have are I've worked out based on guesstimation. So, in my city there's this place called [name of local restaurant] and then make the best quesadillas, but there's absolutely no carb information. So, I just had to work out based on estimation, what carbohydrates to give myself for it. So, if I could save that as like the [name of local restaurant] quesadilla. But then if I was going to McDonald's where that information is widely available online, it can pull in stuff where it is available. So, I guess like a mix it both, so you're utilising as much of what's available as possible. But I'm sure we all have our own brain bank of foods we eat where the carbohydrate information isn't necessarily available.

P12

Yeah, I was going to say it would probably good to have a mix because obviously there's meals that you like, you make up yourself. Like when you're home, make a meal, you're going to have, you're probably gonna have different serving sizes as well because you're not going to put on exactly the same amount every time unless you're weighed out um and things like that. Like something with like, you know, mashed potato, are you're gonna put on exactly this many scoops? Um. Or something like that. Um. But yeah. I think it would be good to have the mix because obviously like you said, things like McDonald's, KFC, even like places like Nando's and things like that, um they've got their own like database bank of um carbohydrate content online or through via different apps. You could pull the information from there and then for your own homemade meals, yeah, you put in the amount of carbs you usually have for it, name it, save it, and then it's just there for the next time you have it. And you can edit it, if your serving size is different, maybe as well.

Interviewer

Yeah. How about you, [P21]? How would she feel about using that as a function on a system?

P21

Yeah, good. I agree with basically what everyone's been saying and I feel like quite often we're creatures of habit and we'll have the same takeaway or the same amount of pasta um on like a weekly basis. Um. So, I think having those kind of options would definitely be useful for most people with a pump.

Interviewer

Yeah. And maybe another point of that, I think like [P03], you were saying around everyone having their own kind of preset of stuff. What. And previously in these focus group people have kind of been encouraging of peer support. In terms of that, what what about if there was kind of a shared nature to it that people were kind of supporting with suggestions for different meals. Would you like that input from others, or do you feel it should be kept more personal to you as an individual?

P12

Sorry you go.

P03

I think I am open to both because I think like like, say, the [local restaurant], I've I'm pretty sure I'm not the only diabetic that goes to this [local restaurant]. It could be really useful for someone else to learn from my experience there and what I eat there. But then things like my own recipes, like my pasta bake probably is going to be different from [P12]'s pasta bake or [P21]'s pasta bake or [P06]'s, or yours. So, I guess it, wouldn't make as much sense to share that. I'd still be open to it, but I think it's about the context of what is being shared, is it a personal recipe or is it just a takeaway restaurant that doesn't have it publicly available.

P12

Yeah, I think if it gave you the option to upload it like. Like you say, if you've gone to like a takeaway, there's no info available, but you've got this certain dish. You won't be the only person to order that specific dish, or even that specific combination of of things. So, if you put that on a database for other people's access, yeah, great. But like, recipes for something that you've made yourself, which is your like, personal preference of flavour and taste and things like that. I mean you could upload it if you wanted to, but it's not something that probably a lot of people are going to go on and be like, "oh yeah, that's exactly my taste or that is something exactly that I would make". And so yeah, I think if it gave you the option, that that would be great, but it's not something that you necessarily have to upload if you don't want to.

P21

I'm just drawing parallels here. I don't know if anyone's used it but MyFitnessPal that has a huge database of um food items sort of packaged goods, but also homemade meals. And you can sort of go on and and see the macros, carbohydrates, fats etcetera for for um sort of McDonald's and things like that, but also meals that people have made themselves. Which I mean I wouldn't necessarily say you you click on it and say, "oh yeah, I'll trust that". But I guess it could give you a good starting point maybe if you didn't know. If you having mac and cheese and so, this persons homemade mac and cheese is this many carbohydrates. And so I think obviously you can rely on it completely but it could be sort of a useful tool.

P12

Yeah, I use an app that I think might be quite similar. Mine's um NutraCheck, which isn't exactly like fitness base, I don't think, but it does have a lot of um information about lots of different foods. Like you say pre packaged ones but also ones from restaurants like McDonald's and things like that. Um. Yeah. And I think you can like, make your own recipes there as well, depending on which ingredients you're use. And again, I I don't know whether that's shared fully online or not, or if that's just personal to you. But yeah, it's it's kind of the same thing, yeah. Can can you trust somebody else's recipe. Because they might use different ingredients to you and it might just be a little bit out and then therefore you get your like your carb counting wrong, your insulin wrong and then you end up

going high or you end up going low. So you could you could trust them or you could not and just stick to your own guns. It's just yeah.

Interviewer

Yeah. I mean, clearly there's a lot of sense in what you're saying if there are companies already working on these um, you know, these apps that fulfil this function. Maybe like [P06], as you mentioned at the beginning, there is this pasta, you have a lot that you've been trying to work out how you deal with it. Can you think for a way of adapting it to maybe be able to help you in your kind of, yeah, understanding of it and the way you manage around it?

P06

So I I yeah, I mean it'd be quite useful to have like. I'm not great, actually, I'd normally just guesstimate most of my meals. And because like I probably have similar amount of pasta each time, I kind of learn off each time, so it does get better. But the thing I struggle with is more that like, what I need to do is give like a second dose of insulin. Um. Which I know some pumps have like functionality where they, you can. I think it's sort of like multiwave I think was on the Accu-Chek I used to have, which work quite well. Whereas, this because I can only do an extended bolus it like kind of doesn't work the same. I kind of need a second hit of a insulin. But I yeah, I probably feel a lot more confident in my uh the amount of like insulin, carbohydrate and putting for them if there was that database online that I could draw from. Um. I just don't know if I'd be good enough to actually check that every time I was making a meal.

Interviewer

Yeah. Yeah, I think that's a very good point and it comes back to, yeah, what we've been discussing already, which is that, you know, it's managing that effort as well that it takes to try and find this extra information.

P03

I guess when you are putting in like your meal or you're in your insulin for the meal, it could say “what are you eating?” And it could automatically pull in suggestions if you say “ohh, I'm having a pastabake”, it can go, “these are cover pasta bakes that are in this database, do any of these look similar?”

Interviewer

Yeah, definitely. What and then you draw your own conclusion between images on them?

P03

Yeah yeah.

Interviewer

Ohh interesting for sure and and I guess that that was work at some point trying to automate that further in terms of you take an image of your food and it tries to estimate the amount of carbohydrate in. I'm not sure how much, I don't think they had much success at the time, but maybe that will be something that can at some point will be possible.

P03

Because I know with Carbs and Cals, they put a picture of, like, pasta on a plate. It it doesn't, well last time I used, yeah, it didn't tell you what the dimensions of the plate were, which was quite annoying. But you could, if you could, like say this is a 30-centimetre plate, this is what the pasta looks like on the plate, it can compare what it looks like to its own pasta on a 30-centimetre plate.

Interviewer

Yeah, definitely. That's an interesting idea. And you kind of build it up that way through, yeah, comparison of images. Oh yeah interesting. Yeah, maybe changing the topic a little bit, another thing I want to talk about um was around sort of, uh, privacy and particularly data. Particularly as I guess we've kind of discussed previously in this study around it's kind of looking at that potential of including data from a smartwatch in a closed loop system, which already has information on blood glucose levels, carbohydrate intake, insulin doses, etcetera. So, you're kind of building up this kind of pile of data there. And I just wondered how people felt about that kind of growing in future, yeah, in, in terms of that privacy level, people potentially gaining access to it.

P12

I definitely think I'd be wary of other people having access to like all my data from my watch. Sure, even all my data from my insulin pump as well. Um. It's like it's like it', that that's personal to me. Nobody else needs to see that, unless I show them it. Like if I want to say “ohh look this is how many this is how many steps I did yesterday”. Other, other than that, nobody needs to know any of the information that's on there. No one needs to know how long I slept or anything like that, anything else that it measures. So, I would be wary if people could access that data. Um. So, as long as the privacy is strong on all these apps and devices and services that we use, then that's fine. But if it's not, then that's where I think there'll be a bit of cause for concern.

Interviewer

Yeah, yeah. What, what about the rest of you, does anyone share that concern? Is anyone less bothered by that information getting out there?

P21

And to be honest, for me, I probably would take quite a different view. Like I don't particularly find the data that sensitive. Um. I mean, obviously I'd rather it not be, you know, one display for the whole world to see. But I'm not, that wouldn't be sort of a big concern for me.

P06

Um. I think I wouldn't really have an issue like other people having access to it as long as it was like anonymous. Um, so kind of like with this study where like our information is getting taken but the end result, you're not going to be able to match it like any of the data to anyone in particular then um yeah, I don't have an issue with that.

P03

Yeah, I think I'd agree if it was anonymous. If it was shared anonymously, I think I'd be fine with it. But if it's like this is [P03]'s 20th of October 2023 and 5:14 PM they were this level and they consumed this many carbs and they gave this much insulin, it feels like it's a bit intrusive.

Interviewer

Yeah, I guess what the. These concerns that some of you have, like expressed around people gaining access to it. What is the? Yeah. What what are those concerns or what are you worried people might do with it?

P03

I guess I'm not worried about what someone's gonna do with the data. It's just, it's me. Um. And I don't know, I don't know why anybody would care about what my blood sugars were at 5:15 on the 20th of October? Outside of like me and my family, who, and friends like seeing my blood sugars just for safety. Like I've chosen to share my Dexcom with my housemate because sometimes he comes and wakes me up in the middle of the night with a glass of milk if I'm low. But yeah, I don't understand why it would need to be shared outside of that. Like what would um Joe Bloggs down the street do with my data? Why does he care about that? And I guess if it gets seen by people at work or managers at work, they could you maybe use against you at times. Like if I was having a hypo at work and I'd want to go take like 20 minutes to go sit down. Or I wasn't quite hypo, but I was like. 4.4, going down, had a crash and I was like “well, I'm going to go low”. They’d be like, “well, actually, you're only 4.4, you're not quite low yet”. Whereas, at the minute it's on a trust basis.

Interviewer

Yeah, I I think that's an interesting example. I I guess a similar one I've heard as well as around insurance companies as well and if they had access to those readings that gave yeah, an overview of what your diabetes management was at different times, whether that would affect premiums, you'd have to pay for different insurances. Yeah and I guess that, yeah, yeah. A few people raised this around that there are bits that you already share with certain people in terms of maybe blood

glucose readings with friends and family or the data to me in this study. I wonder has that changed over time? Are there times you you wouldn't want to give any of your data away or are there times where you'd be happy to give more of it away?

P12

I think maybe if I actually had the option, because I don't think I don't think the Medtronic or the Guardian actually give you the option to share your levels with anyone or anything like that. But possibly if I could maybe maybe when I'm at work, um to one manager that I might actually trust with it. Like there's certain ones where, like, no, I'd rather you didn't have the access to be honest. But there's other ones where I'm like, yes, I know you would like check on it like very quickly, see if I'm alright, yeah, you're fine, you just like we'll ignore it. We'll keep going and like, if it's starting to go low, they'll come up and be like like, “are you OK?” Yeah, like make sure I'm alright, things like that. But I think other than that I'm I wouldn't be sharing it very often with very many people. Um. Just because they don't need to know what my blood sugars doing at any given time, unless, like unless it was like a dangerous kind of low level. They don't really need to know. I can tell them, if they need to know. Just they don't need to be checking up on me all the time. I feel like that's that's a little bit of an intrusion and I know it would be because I'm sharing it but yeah, if they're looking all the time and just try to like keep tab on it it's personally would be a bit of an intrusion for me and like overstepping a little bit. And yeah, I'm I'm I'm just a bit funny with what I share with people personally, which is why I think the privacy issue is probably more of an issue with me than it is with everybody else, from what I've heard there. But yeah.

Interviewer

Yeah, definitely. I think it varies a lot of different things. I mean obviously there'd be some things that maybe change that a little more. For example, yeah, so talking a smart watch coming in and suddenly they're being GPS data stored, so people could be aware of where you are or you know where you commonly travel, if suddenly it's being stored somewhere. Um. And that really varies over time. The yeah. Yeah, I think just the, the final thing maybe to go over is like similar to previous focus groups, it's just more reviewing that last month of using the Fitbit or whichever smart which you've chosen to use. Um. Yeah, and just if anything has kind of changed in the last month in terms of the way you're using it. If there's a particular feature using more, if you think generally you've been using it more or less, anything. Yeah, around that would be interesting, if anyone's got anything to share?

P12

I I went on holiday last month I went to Tenerife for 10 days. Um. And obviously I wasn't wearing it when I was in the sun a lot or by the pool or anything, obviously just in case it got wet or it overheated and then stopped working. I didn't want any of that to happen, so I left it up in the room. But when we were going, when we went out on a night I would put it on. And I left it on overnight for sleep and things like that. And and I still managed to get like decent amount of steps in, is what I realised. We were just walking a lot to get to like restaurants and then to bars and stuff like that. And it it just kind of shows you how much you walk when you're away on holidays like that. Like just walking from place to place to place on a night. Which I don't think I've ever really thought of before, but then I've got the proof from the watch data, which is quite which was quite interesting to see.

Like even though I wasn't wearing it for like half of the day I was still getting nearly 10,000 steps in, just because of how much walking we did on a night. And that was quite interesting to see.

Interviewer

Yeah. Did that trigger a change um as a result of saying that ever?

P12

I think I'll just try walking a little bit more, like just in daily life. Like places where I could have just driven to, I'll start like it, would have only been a short distance. Oh, no, I'll just, I'll walk it instead, if the weather's not like it's been today, which is torrentia. Um. But yeah, just started making me think “ohh yeah, I could walk that a little bit more”. It's um just because you don't think about how, when, when you're on like a holiday like that, you don't really think of like the distance you're walking or how long it'll take you to get to here, or to here. So why should that stop when you get home?

Interviewer

Yeah. Has anyone noticed similar that kind of triggered awareness of the activity they're doing because of the the wearable?

P06

Um. I actually had like a week where I was hitting 10,000 every day and I just decided to up it because it would be interesting to see if I had upped it, whether I'd like then try and walk more than I did. But I'm not sure I did. I just kind of stopped noticing it or like forgot that I'd even changed it and kind of became less aware about it. And also, similarly like I went, I went on a cycle the a couple of weeks ago and I thought I'd try the like fitness function on it and I switched it on and then just totally forgot that I'd, uh turned it on. So, it wasn't even like until late in the evening that I'd switched off. So yeah, I wasn't overly sold on that. Maybe if you picked it up, it becomes bit more habitual and then you remember, but uh.

Interviewer

Yeah. Is it something you'd likely use again? Do you think after that or?

P06

Probably not. I mean, even like I didn't really even look at the data. I just I wasn't that interested in it, um, despite having it.

P12

Funny you said about how you forgot to turn it off? When I take my dog out for a walk, the watch automatically starts like a walk like workout. Not like a workout but like an activity track. Um. Like it will automatically start it after however many minutes. Like my dog walks slow, so it takes us longer to get around the field and get back home so it automatically triggers that. And then there's times where I'll stop it when we get to the front door, because obviously walks over. And then there's times where if I'm trying to like guide her into the house properly, because she's blind and she can't see. I'll try and guide her in and I'll try and like I'll have to dry her feet off and things like that and I'll forget to turn the walk off. But then I'll get it like it'll flash up on the screen saying like, “there's been barely any detective, movement detected for a little bit, is your walk over?” So if it was a one like that for like cycling, or different kinds of activities like the running and things like that, maybe that might make people use it more. Because they wouldn't have even if they forget to turn it off, they would get reminded to turn it off because obviously once they've stopped moving as much the watch you can tell that the workouts over then.

P06

Yeah, that's quite good idea. I did notice that because it noticed my heart rate had, like, gone back down. So, from that I was able to tell when I'd actually stopped uh yeah exercising.

Interviewer

Yeah. Anyone else thought anything interesting of that last month of using it? No, that sounds good.